Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

F. 5 amps, 2.4 ohms

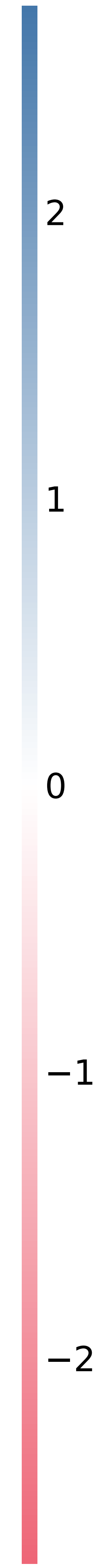
G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms

Reasoning + Answer:



<think>

1. Identify the given information:

- Voltage (V) = 12 volts

- Power (P) = 40 watts

Formula: Power (P) = Voltage (V) x Current (I) x Resistance (R)

We can rearrange this formula to find Current (I) = P / (V x R)

2. Rearrange the formula to find Resistance (R) = V x I / P

3. Use the given options to find the correct combination of current (I) and resistance (R) that satisfies the formula P = V x I x R.

4. After calculating, I found that R = 12 / 40 = 0.3. Now we determine the current that it draws when given all of its answer options and calculate the resistance